Agentic AI-Based Model Regeneration Framework for O-RAN

Ramanujam P
[Department], [Institution]
[City], [Country]
pramanujam68@gmail.com

Nitesh N D
[Department], [Institution]
[City], [Country]
niteshdwaraka@gmail.com

Tamilselvan R
[Department], [Institution]
[City], [Country]
tamilraj1402@gmail.com

Bhuvaneshwar T S
[Department], [Institution]
[City], [Country]
bhuvaneshwarts1@gmail.com

***Abstract*—AI/ML models deployed in the Open Radio Access Network (O-RAN) degrade as traffic, load, and radio conditions evolve, and current MLOps workflows leave the choice of remedy to human operators. This paper presents an Agentic AI-based model regeneration framework that acts on a reported degradation and decides, without human intervention, how the deployed model should be readapted. After quantifying the shift with the Kolmogorov–Smirnov test and the Population Stability Index (PSI), the framework attempts a *similar model exchange* that re-scores every registered version on held-out current data and promotes one that measurably beats the deployed model. Otherwise a Decision Engine, combining hard constraints with an optional large language model (LLM) agent that calls read-only evidence tools, selects Finetune, Regenerate, Rollback, or No Action, and an engine selection stage runs the action with a native engine or, only if none exists, with LLM-written code that is checked against an abstract-syntax-tree whitelist and sandboxed. A paired-bootstrap gate admits a candidate only when its improvement is established. On eight controlled drift scenarios, remediation reduces held-out RMSE by 62.5–97.5 % under concept drift but changes it by less than 1 % under pure covariate shift although PSI reaches 15.6, so PSI is a sound severity measure but an unsafe trigger. The gate rejects a candidate whose 95 % interval straddles zero although its point estimate exceeds in magnitude that of an accepted candidate.**

***Index Terms*—Agentic AI, O-RAN, MLOps, model lifecycle management, concept drift, population stability index, bootstrap, large language model.**

I. Introduction

The Open Radio Access Network (O-RAN) enables intelligent, programmable RAN operation through software-defined control and AI/ML applications hosted in the RAN Intelligent Controllers [1]. Models that run continuously in such an environment degrade as network behaviour, workload, and operating conditions change: the input distribution $P(X)$, the input–output relationship $P(y \mid X)$, or both shift over time [2].

Existing MLOps workflows for O-RAN monitor models and retrain them through hand-built pipelines [3], [4], but the decision of *how* to remediate a degraded production model, by reusing an earlier version, fine-tuning, retraining from scratch, rolling back, or doing nothing, remains with a human operator. This paper addresses that decision. Its contributions are:

- a closed-loop readaptation pipeline that measures the shift, attempts a similar model exchange from the version registry, and otherwise lets a constraint-bounded Decision Engine choose among Finetune, Regenerate, Rollback, and No Action;
- an engine selection stage that falls back to LLM-written adaptation code only when no native engine exists, with an AST whitelist, sandboxed execution, and a bounded write–check–run–fix loop;
- a paired-bootstrap deployment gate that admits a candidate only if its improvement is established at 95 % confidence; and
- a reproducible evaluation on eight controlled drift scenarios, showing why PSI belongs after a degradation report as a severity measure rather than before it as a trigger.

II. Related Work

Table I summarises the fifteen works reviewed, in four groups.

Agentic and closed-loop control. Mekrache *et al.* [5] place an LLM at the centre of the intent life cycle on a 5G testbed, but do not decide when a deployed model must be retrained. DRL xApps run in O-RAN control loops [6], and ColO-RAN [7] shows that they need online retraining once deployed, leaving when and how to the designer.

Lifecycle management. RLOps [3] lays out an MLOps life cycle for learning-based O-RAN control without automating the choice of action. REASON [4] adds AI orchestration, monitoring, and a digital-twin check before deployment, but its pipelines are not driven by a remediation decision.

Conflict and intent. Adamczyk and Kliks [8] detect direct, indirect, and implicit xApp conflicts in the near-RT RIC, and QACM [9] mitigates them with QoS awareness; surveys cover LLMs for telecommunications [10] and intent-based networking [11]. None manages a model's lifecycle after it degrades.

Twins, orchestration, and drift. Digital twins test changes off the live network [12] and gain a 6G reference architecture [13]; a DRL traffic-steering xApp raises throughput by 50 % [14]; OrchestRAN [15] decides which models run where; and a drift module for the 5G core detects drift and retrains [16]. The remediation decision rests on the concept-drift literature: Ditzler *et al.* [2] separate covariate shift, where $P(X)$ changes, from real concept drift, where $P(y \mid X)$ changes, and Lu *et al.* [17] survey detection and adaptation. Neither prescribes

which remedy fits a given shift signature, which is the decision the Decision Engine makes.

III. Research Gaps

The reviewed works leave three gaps that this framework addresses.

- *G1 (Autonomous readaptation):* when to retrain and which action to take remain open [5]–[7], and life cycles stop at hand-built pipelines [3], [4]. An autonomous readapter should first try to reuse an earlier model version that already fits the current data.
- *G2 (Remediation logic):* a degradation signal is uncertain—xApp conflicts can degrade a model that is still correct [8]—and drift modules [16] retrain without choosing a remedy. Missing is a Finetune, Regenerate, Rollback, or No Action decision that acts on a reported degradation and uses the measured shift only as severity.
- *G3 (Verified promotion):* trust gates, twins, and model orchestration [4], [13], [15] decide what is deployed, but none states a statistical rule that rejects a candidate whose improvement is not established.

IV. Proposed Framework

A. Pipeline Overview

Fig. 1 and Algorithm 1 give the pipeline end to end. The framework does not detect drift itself: an external monitor, or one of the framework's declarative event mappers, reports a degradation for a model and names the current data. Each report becomes a job that is validated, persisted, and queued; a separate worker executes it through the stages below, and every transition of its sixteen-state life cycle is recorded with a timestamp, actor, and reason.

B. Model Intake

The Intake Agent validates the report, checks that the model exists in the MLflow registry, and enqueues the job; it never runs the pipeline in the API process. A worker leases the job, so a crashed worker's job is picked up again, and an optional per-job deadline ends overdue jobs as timed out.

C. Shift Analysis

The current data and the deployed version's training data are retrieved from versioned, content-hashed data versions whose lineage links each model version to the exact rows it was trained on. For every numeric feature the framework computes the two-sample Kolmogorov–Smirnov test and the PSI of Eq. (1). These statistics are evidence for the later stages; they do not by themselves start or stop remediation.

D. Similar Model Exchange

Rather than a separate catalog of models and environments, the version registry itself is searched. Up to ten of the newest registered versions, and the deployed one, are re-scored on the held-out slice H of the current data. A version is compatible when its artifact passes its checksum, loads, is of the same estimator kind, and finds all its features in the data. A compatible, non-deployed version that lowers the deployed version's RMSE by at least 5 % (or raises a classifier's accuracy by at least 0.02) is promoted directly, with no training and no candidate gate. Otherwise the scores are handed to the Decision Engine.

Algorithm 1 Readaptation pipeline for one degradation report

Input: model ID, reference to the current data D_c

Validate the report; reject it if the schema check fails or the model is unknown
Load the training data D_r of the deployed version M_t and the current data D_c
Compute the KS p-value and PSI (Eq. 1) of every feature
Hold out the newest fraction H of D_c (never trained on)
Score every registered version on H (similar model exchange)
if a compatible version beats M_t by the reuse margin **then**
    Promote that version; **return**
end if
$a \leftarrow$ Decision Engine (Eqs. 2–3)
if $a \in \{\text{Rollback}, \text{No Action}\}$ **then**
    Retain M_t; **return**
end if
$M_{\text{cand}} \leftarrow$ native engine of the model type for a; if none, a full retrain when supported, else LLM-written code (AST check, sandbox, bounded repair loop)
if the gate of Eq. 4 accepts M_{cand} on H **then**
    Register M_{cand} and roll it out with the configured delivery strategy
else
    Retain M_t
end if

E. Decision Engine

The Decision Engine chooses among the actions of Eq. (2). Hard constraints come first and cannot be overruled: too few current rows or an insufficient sample stop the job as *insufficient information*; a framework without an adaptation engine stops it as *no compatible strategy*; a maximum feature PSI of at least $\tau = 0.5$ rules out Finetune, so a severe shift is always regenerated; and Rollback requires recent performance data and a compatible earlier version. When an LLM is configured, it chooses among the remaining actions, either from a single structured prompt or, if enabled, as an agent that calls read-only tools (shift summary, per-feature statistics, version scores, constraints, per-action cost and expected-gain estimates) for a bounded number of steps before it submits. An answer outside the feasible set, a malformed answer, an unreachable LLM, or an exhausted step budget falls back to the deterministic preference order of Eq. (3). Every decision is stored with its evidence, thresholds, source, and, for the agent, its full tool trace. A Rollback or No Action decision trains nothing and keeps the deployed version; re-promoting an earlier version is the job of the similar model exchange, which acts only when that version measurably wins.

TABLE I
SUMMARY OF THE REVIEWED LITERATURE

No.	Work	Venue, year	Category	Key points and gap with respect to this work
1	LLM-centric intent management [5]	IEEE Network, 2024	Agentic; intent; LLM	LLM translates and assures intents; no retraining decision
2	Intelligence in O-RAN [6]	IEEE Commun. Mag., 2021	Closed-loop AI control	DRL xApps in RIC control loops; model upkeep left open
3	ColO-RAN [7]	IEEE Trans. Mobile Comput., 2023	ML xApp pipeline	Deployed agents need online retraining; when and how not decided
4	RLOps [3]	IEEE Access, 2022	ML life cycle (MLOps)	Specify, develop, serve, monitor, retrain; action choice is manual
5	REASON [4]	IEEE Wireless Commun., 2025	Trustworthy AI lifecycle	Orchestration, monitoring, twin check before deployment; no remedy choice
6	Near-RT RIC conflicts [8]	IEEE Commun. Mag., 2023	xApp conflict detection	Direct, indirect, implicit conflicts; no model lifecycle
7	LLM for telecommunications [10]	IEEE Commun. Surveys Tuts., 2025	LLM survey	Generation, classification, optimisation; no model readaptation
8	Intent-based networking [11]	IEEE Commun. Surveys Tuts., 2023	Intent-based networking	Translation, activation, assurance; no ML lifecycle
9	QACM [9]	IEEE Trans. Green Commun. Netw., 2024	xApp conflict mitigation	Sets conflicting parameters so most xApps meet QoS
10	Digital twin for Open RAN [12]	IEEE Commun. Mag., 2023	Network digital twin	Twin tests changes off the live network; use cases
11	Traffic-steering xApp [14]	IEEE Trans. Mobile Comput., 2024	DRL handover control	50 % higher throughput; trained once, no retraining policy
12	Reference NDT architecture [13]	IEEE Open J. Commun. Soc., 2026	Digital twin architecture	AI workflows, simulation, data, orchestration, twin lifecycle
13	OrchestRAN [15]	IEEE INFOCOM, 2022	Model orchestration	Chooses which models run where; no degradation-driven update
14	Model drift in 5G core [16]	IEEE MeditCom, 2022	Drift detection and adaptation	Detects drift and retrains; a single remedy, no choice
15	Understanding O-RAN [1]	IEEE Commun. Surveys Tuts., 2023	O-RAN architecture	Architecture, interfaces, algorithms, security, challenges

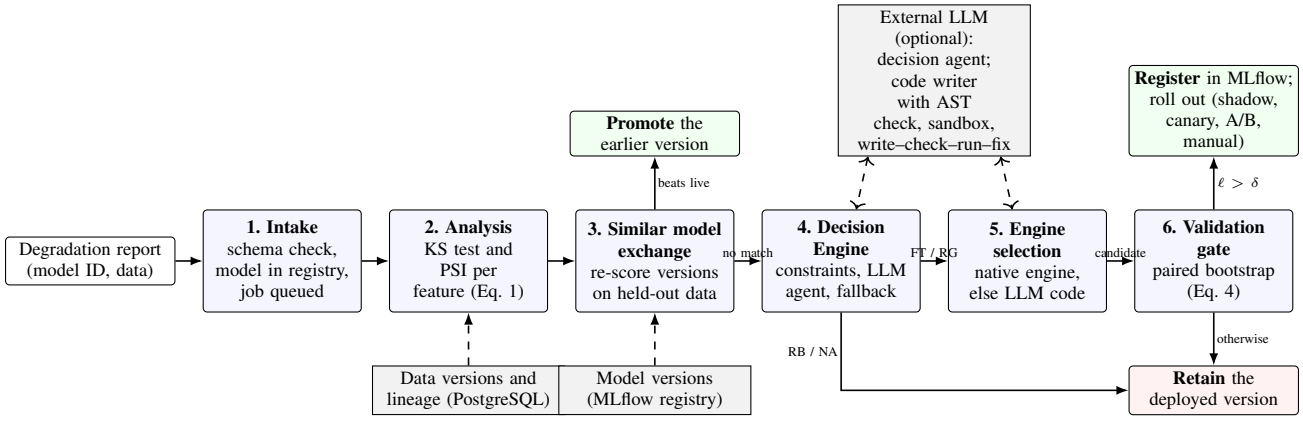


Fig. 1. Architecture of the regeneration pipeline. Solid arrows: control flow; dashed arrows: data and optional LLM calls. FT: Finetune; RG: Regenerate; RB: Rollback; NA: No Action.

F. Engine Selection

Finetune continues training a copy of the deployed model from its current state: through `partial_fit` for incremental scikit-learn estimators and by continued gradient descent from the current weights for PyTorch modules. Regenerate refits the same estimator configuration from scratch. Each model family is a registered model-type plugin (scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch, PyTorch sequence models, Keras, ONNX, and statsmodels), so a new family is added by registration rather than by changing the pipeline. If the plugin has no engine for the chosen action, a Finetune falls back to a full retrain when the model supports it; only otherwise, and only when an LLM is configured, an external LLM writes an `adapt()` function. That code is rejected unless it passes an AST whitelist of imports, built-ins, and attributes, and it runs in a sandboxed subprocess or container with time and memory limits; only the fitted model crosses back. Optionally, a rejected or failing attempt is returned to the LLM with the scan violations or the sandbox traceback for a bounded number of repair attempts. If every path fails, the job ends with a typed error and the deployed version is retained.

G. Validation, Registration, and Delivery

The candidate is scored against the deployed model on the same held-out slice H by the gate of Eq. (4), together with configurable guardrails on data slices, calibration, latency, and model size. An accepted candidate is registered as a new MLflow version with the data version it was trained on, and rolled out by the configured delivery strategy (shadow, canary, blue–green, A/B, or manual approval) with automatic rollback on a breached health rule.

V. Mathematical Formulation

A. Shift Severity

For a feature with reference sample (training data) and current sample, the PSI over B bins is

$$\mathrm{PSI} = \sum_{i=1}^{B} (P_i - Q_i)\, \ln \frac{P_i}{Q_i}, \tag{1}$$

where P_i and Q_i are the reference and current proportions in bin i. The bins follow Sturges' rule over the pooled sample, and an empty bin is assigned a proportion of 10^{-4}, so a shift that moves the current data outside the reference range yields large values (above 15 in Section VIII). The pipeline uses $\mathrm{PSI}_{\max}$, the maximum over features.

B. Readaptation Policy

The action set is

$$\mathcal{A} = \{\text{Finetune}, \text{Regenerate}, \text{Rollback}, \text{No Action}\}. \tag{2}$$

Let $\mathcal{R}(c) \subseteq \mathcal{A}$ be the actions ruled out by the hard constraints c, including Finetune whenever $\mathrm{PSI}_{\max} \geq \tau$. With the feasible set $\mathcal{F} = \mathcal{A} \setminus \mathcal{R}(c)$, the selected action is

$$a^* = \begin{cases} a_{\mathrm{LLM}}, & \text{if } a_{\mathrm{LLM}} \in \mathcal{F}, \\ \text{first of } \mathcal{F} \text{ in } (\mathrm{FT}, \mathrm{RG}, \mathrm{RB}, \mathrm{NA}), & \text{otherwise}, \end{cases} \tag{3}$$

where a_{LLM} is the LLM's answer, if any, and the fallback prefers the least costly feasible action.

C. Deployment Gate Criterion

A candidate produced by Finetune or Regenerate is admitted only if it is demonstrably superior to the incumbent, not merely different from it. Let m be the quality metric oriented so that larger is better, and let H be the time-ordered held-out slice on which neither model was trained. Drawing B' paired bootstrap resamples [18], both models are scored on the same resampled rows, and the improvement $\Delta = m(M_{\text{cand}}) - m(M_t)$ receives a two-sided percentile interval $[\ell, u]$ at confidence $1 - \alpha$. The candidate is deployed when

$$\ell > \delta, \qquad \delta = 0,\ 1 - \alpha = 0.95,\ B' = 1000. \tag{4}$$

Because the criterion depends on the lower bound ℓ rather than on the point estimate Δ, any candidate whose interval contains zero is rejected.

VI. Implementation

The framework comprises 33 528 lines of Python in 186 modules, with 961 automated tests. The service exposes 39 REST endpoints covering intake, job control, model and data versions, gate decisions, and rollouts. Intake only validates and enqueues; separate workers lease jobs from a durable queue (database, Celery, RQ, or Kubernetes jobs). The data versions and the job, audit, and gate records are kept in PostgreSQL (SQLite for development), and model artifacts in MLflow. The LLM is reached through one port with interchangeable providers (Google Gemini by default, Anthropic, or any OpenAI-compatible endpoint, including a self-hosted one); every call is recorded with its versioned prompt, tokens, and cost, guarded by retries, a circuit breaker, and a budget, and without a configured key every decision takes the deterministic path.

TABLE II
Drift Scenarios and the Measured Shift Severity

ID	S0	S1	S2	S3	S4	S5	S6	S7
Shift	none	cov.	cov.	conc.	conc.	mixed	mixed	severe
Strength	–	mild	strong	mild	strong	mild	strong	–
Mean shift	0	0.2	0.6	0	0	0.2	0.6	0.9
β	1.0	1.0	1.0	1.5	2.5	1.5	2.5	4.0
$P(y \mid X)$ changed	no	no	no	yes	yes	yes	yes	yes
$\mathrm{PSI}_{\max}$	0.058	4.949	15.635	0.058	0.103	5.283	15.638	16.090

TABLE III
Held-Out RMSE by Scenario and Remediation Action

Action	S0	S1	S2	S3	S4	S5	S6	S7
No remed.	0.0915	0.1037	0.0890	0.2542	0.7697	0.3587	1.6788	4.2301
Finetune	0.0965	**0.1032**	**0.0884**	0.1009	0.1613	0.1131	0.1402	0.1955
Regenerate	**0.0936**	0.1059	0.0908	**0.0953**	**0.0920**	**0.1101**	**0.1069**	**0.1056**
Gain (%)	−2.29	0.52	0.63	62.49	88.05	69.29	93.63	97.50

Gain: RMSE reduction of the better remediation arm (bold) relative to no remediation.

VII. Experimental Setup

The evaluation uses synthetic cell-level KPIs with three features, PRB utilisation (`prb_util`), CQI (`cqi`), and RSRP (`rsrp`), scaled to about $[0, 1]$, against throughput $y = \beta\,\mathtt{prb_util} + 0.8\,\mathtt{cqi} + 0.5\,\mathtt{rsrp} + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.1^2)$. A baseline window of 600 records represents the environment of the deployed model M_t, an SGD linear regressor. Eight drifted windows of 600 records vary two generator parameters independently (Table II): the mean of `prb_util`, which alters $P(X)$ only, and its coefficient β (1.0 in the baseline), which alters $P(y \mid X)$. Only the second genuinely invalidates M_t.

Each drifted window is split in time order; the newest 30 % is held out and never used for training, since random splitting would leak future information. Three arms are compared on the identical held-out slice: (i) No remediation, the counterfactual in which M_t keeps serving; (ii) Finetune, one incremental `partial_fit` pass seeded from M_t; and (iii) Regenerate, a full refit of the same estimator configuration. Both remediation arms call the framework's own scikit-learn engines, PSI is computed by the analysis stage's code, and the gate is the framework's validation engine with its default policy. No LLM key was configured, so every decision took the deterministic path. All windows use fixed seeds, and the script `scripts/paper_experiments.py` in the project repository regenerates every number and figure in Section VIII.

VIII. Results and Analysis

A. Effectiveness of the Remediation Actions

Table III and Fig. 2 report held-out RMSE for each arm. Where the conditional relationship changed (S3–S7), remediation reduces RMSE by 62.49–97.50 %. Regenerate outperforms Finetune in every such scenario, and the margin widens

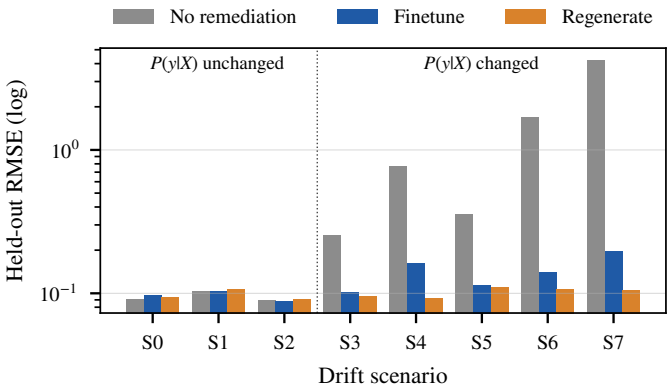


Fig. 2. Held-out RMSE by scenario and remediation action (logarithmic axis). Regenerate is the stronger action wherever $P(y \mid X)$ has changed.

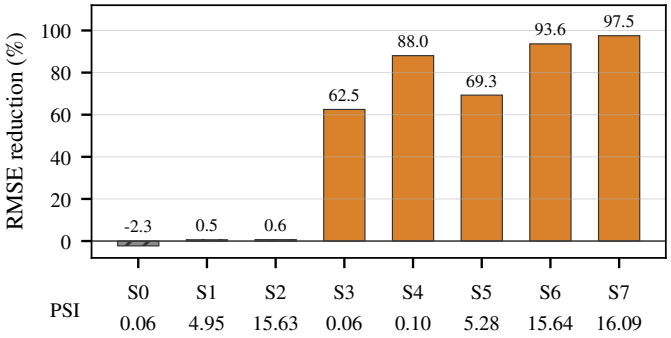


Fig. 3. RMSE reduction by the better remediation arm per scenario, with $\mathrm{PSI}_{\max}$ under each label. High PSI without concept drift (hatched) brings no benefit; low PSI with concept drift (S3) brings a large one.

with severity: at S7, Regenerate reaches 0.1056 against 0.1955 for Finetune, 46.0 % lower, because a single incremental pass from the stale weights cannot absorb a fourfold change of the coefficient. Finetune remains the cheaper choice where the shift is mild and the evidence does not demand a full regeneration.

Where only the input distribution moved (S1–S2), remediation changes RMSE by less than 1 %, and with no shift at all (S0) the better arm is 2.29 % worse than the deployed model. This is the expected behaviour of a correctly specified model: shifting $P(X)$ without altering $P(y \mid X)$ leaves the learned mapping valid, and refitting on a smaller window only adds estimation variance. Here the pipeline should retain the deployed model rather than spend computation on regeneration, which is what the gate enforces: a candidate whose improvement is not established is not deployed.

B. PSI as a Severity Measure, Not a Trigger

Fig. 3 shows the benefit of remediation per scenario with each scenario's $\mathrm{PSI}_{\max}$. Scenario S2 records a PSI of 15.6, far above any conventional alert threshold (0.25), yet remediation improves RMSE by only 0.63 %. Scenario S3 records 0.058, below every conventional threshold, yet remediation improves RMSE by 62.49 %. This supports the design of Fig. 1: PSI is computed after a degradation has been reported and only shapes the action (ruling out Finetune at $\mathrm{PSI}_{\max} \geq 0.5$). As an entry condition it would have regenerated in S2, where nothing is gained, and stayed idle in S3, where the model had become invalid; PSI must never trigger remediation on its own.

TABLE IV
DEPLOYMENT GATE DECISIONS (95 % PAIRED BOOTSTRAP, 1000 RESAMPLES)

Candidate	Δ	ℓ	u	>0 (%)	Verdict
Exact least squares	$+0.00371$	$+0.00037$	$+0.00724$	98.9	Deploy
SGD, other seed	$+0.00084$	$+0.00034$	$+0.00137$	100.0	Deploy
Boosted stumps	-0.00390	-0.00956	$+0.00186$	10.0	Reject
Copy of incumbent	0.00000	0.00000	0.00000	0.0	Reject
Stale (baseline fit)	-0.25842	-0.27477	-0.24131	0.0	Reject
Constant (mean)	-0.11486	-0.13934	-0.09204	0.0	Reject

Δ: RMSE reduction against the incumbent (RMSE 0.1008); >0: share of resamples with $\Delta > 0$.

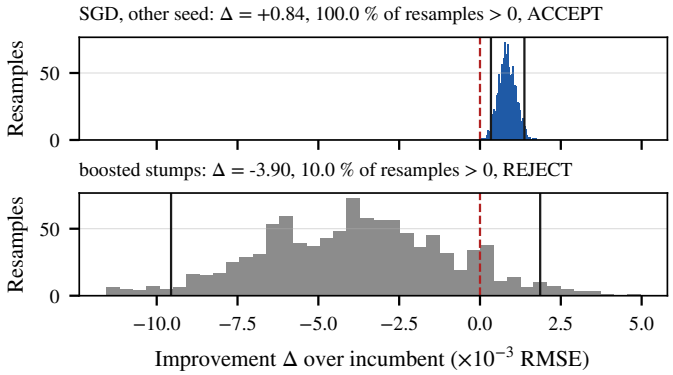


Fig. 4. Bootstrap distributions of the two decisive candidates. Solid lines: 95 % interval; dashed line: $\delta = 0$.

C. Deployment Gate Discrimination

The gate of Eq. (4) was exercised against six candidates of controlled quality. The incumbent was itself fitted to a drifted window drawn with the parameters of S5, so all differences are small; against a stale incumbent every candidate would look excellent. Table IV and Fig. 4 report the outcome.

The second and third rows are decisive. A second SGD fit with another seed improves RMSE by only 0.00084, yet its entire interval lies above zero (100 % of resamples), so it is deployed. A model of 300 boosted decision stumps has a point estimate four times larger in magnitude, -0.00390, but its interval spans zero (10.0 % of resamples above it): it is not shown to differ from the incumbent, and a lenient non-inferiority check on point estimates with a tolerance of 0.004 would have promoted it. Under Eq. (4) it is rejected and the incumbent is retained. An identical copy ($\Delta = 0$ exactly) is rejected as well, because $\ell > 0$ is strict.

D. Computational Cost

Fig. 5 gives the per-stage wall time of one Regenerate run, averaged over five repetitions after a warm-up run, about 385 ms in total. The shift analysis (213.1 ± 19.5 ms), which

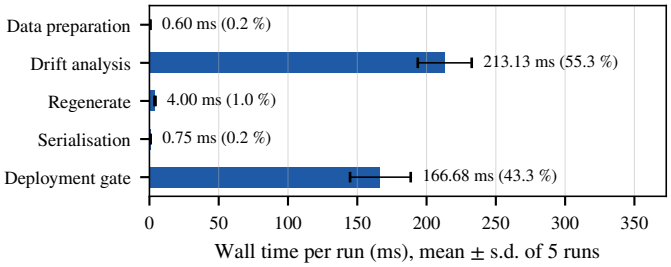


Fig. 5. Wall time of one Regenerate run per stage (4 vCPU, Intel Xeon 2.1 GHz).

builds an Evidently report for the KS test and PSI of every feature, and the gate (166.7 ± 21.9 ms) for its 1000 paired resamples dominate; the refit itself takes 4 ms on this small linear model. Both costs are negligible against the cadence of readaptation requests. A similar model exchange skips the training and the candidate gate, but not the shift analysis or the scoring of the registered versions.

IX. Limitations

The similar model exchange path runs end to end, but its benefit was not quantified; that needs a registry with many recorded versions and environments. The results use synthetic KPIs with controlled covariate and concept shift and a linear model, not an operator trace, and the remediation arms were executed with the scikit-learn engines only. Fine-tuning is a warm-start continuation; parameter-efficient variants such as low-rank adaptation or partial layer freezing, which limit catastrophic forgetting [19], are not implemented. A Rollback decision of the Decision Engine does not itself re-promote a version. The framework treats a reported degradation as drift; a degradation caused by conflicting xApps [8], [9], which calls for mitigation rather than retraining, is not diagnosed. Neither the LLM decision agent nor the LLM-written engines and their repair loop were evaluated against a live LLM; they were tested with a scripted LLM. Finally, registries, serving systems, and brokers were exercised against emulators rather than live managed services.

X. Conclusion and Future Work

This paper presented an Agentic AI-based model regeneration framework for O-RAN that addresses a question the reviewed work leaves open: when, and how, to readapt a degraded model. After measuring the shift, a readapter first attempts a similar model exchange from the version registry; otherwise a constraint-bounded Decision Engine selects Fine-tune, Regenerate, Rollback, or No Action, and the engine selection stage runs it, writing code with an LLM only when no native engine exists.

The measured results support three claims. Regeneration is the stronger remediation action under genuine concept drift, improving held-out RMSE by up to 97.5 %. Remediation brings no material benefit under pure covariate shift even at a PSI of 15.6, which shows that PSI belongs after a reported degradation as a severity measure, not before it as a trigger. And the deployment gate rejects a candidate whose improvement interval includes zero even though its point estimate is larger in magnitude than that of a deployed candidate.

Future work will measure the similar model exchange under realistic O-RAN shift, evaluate the LLM decision agent and the LLM-written engines against live models and oracle choices on recorded operator drift events, add parameter-efficient fine-tuning for deep-learning models, validate candidates in a digital twin before the gate [12], [13], and diagnose xApp conflicts separately from drift.

References

[1] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "Understanding O-RAN: Architecture, interfaces, algorithms, security, and research challenges," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 1376–1411, 2023.

[2] G. Ditzler, M. Roveri, C. Alippi, and R. Polikar, "Learning in nonstationary environments: A survey," *IEEE Comput. Intell. Mag.*, vol. 10, no. 4, pp. 12–25, 2015.

[3] P. Li *et al.*, "RLOps: Development life-cycle of reinforcement learning aided open RAN," *IEEE Access*, vol. 10, pp. 113808–113826, 2022.

[4] J. Parra-Ullauri *et al.*, "Lifecycle management of trustworthy AI models in 6G networks: The REASON approach," *IEEE Wireless Commun.*, vol. 32, no. 2, pp. 42–51, 2025.

[5] A. Mekrache, A. Ksentini, and C. Verikoukis, "Intent-based management of next-generation networks: An LLM-centric approach," *IEEE Netw.*, vol. 38, no. 5, pp. 29–36, 2024.

[6] L. Bonati, S. D'Oro, M. Polese, S. Basagni, and T. Melodia, "Intelligence and learning in O-RAN for data-driven NextG cellular networks," *IEEE Commun. Mag.*, vol. 59, no. 10, pp. 21–27, 2021.

[7] M. Polese, L. Bonati, S. D'Oro, S. Basagni, and T. Melodia, "ColO-RAN: Developing machine learning-based xApps for open RAN closed-loop control on programmable experimental platforms," *IEEE Trans. Mobile Comput.*, vol. 22, no. 10, pp. 5787–5800, 2023.

[8] C. Adamczyk and A. Kliks, "Conflict mitigation framework and conflict detection in O-RAN near-RT RIC," *IEEE Commun. Mag.*, vol. 61, no. 12, pp. 199–205, 2023.

[9] A. Wadud, F. Golpayegani, and N. Afraz, "QACM: QoS-aware xApp conflict mitigation in open RAN," *IEEE Trans. Green Commun. Netw.*, vol. 8, no. 3, pp. 978–993, 2024.

[10] H. Zhou *et al.*, "Large language model (LLM) for telecommunications: A comprehensive survey on principles, key techniques, and opportunities," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 3, pp. 1955–2005, 2025.

[11] A. Leivadeas and M. Falkner, "A survey on intent-based networking," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 625–655, 2023.

[12] A. Masaracchia, V. Sharma, M. Fahim, O. A. Dobre, and T. Q. Duong, "Digital twin for open RAN: Toward intelligent and resilient 6G radio access networks," *IEEE Commun. Mag.*, vol. 61, no. 11, pp. 112–118, 2023.

[13] A. Zaki-Hindi *et al.*, "A reference functional architecture for network digital twins in 6G systems," *IEEE Open J. Commun. Soc.*, vol. 7, pp. 2068–2101, 2026.

[14] A. Lacava *et al.*, "Programmable and customized intelligence for traffic steering in 5G networks using open RAN architectures," *IEEE Trans. Mobile Comput.*, vol. 23, no. 4, pp. 2882–2897, 2024.

[15] S. D'Oro, L. Bonati, M. Polese, and T. Melodia, "OrchestRAN: Network automation through orchestrated intelligence in the open RAN," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, 2022, pp. 270–279.

[16] D. M. Manias, A. Chouman, and A. Shami, "A model drift detection and adaptation framework for 5G core networks," in *Proc. IEEE Int. Mediterranean Conf. Commun. Netw. (MeditCom)*, 2022, pp. 197–202.

[17] J. Lu, A. Liu, F. Dong, F. Gu, J. Gama, and G. Zhang, "Learning under concept drift: A review," *IEEE Trans. Knowl. Data Eng.*, vol. 31, no. 12, pp. 2346–2363, 2019.

[18] A. M. Zoubir and B. Boashash, "The bootstrap and its application in signal processing," *IEEE Signal Process. Mag.*, vol. 15, no. 1, pp. 56–76, 1998.

[19] Y. Kong, L. Liu, H. Chen, J. Kacprzyk, and D. Tao, "Overcoming catastrophic forgetting in continual learning by exploring eigenvalues of Hessian matrix," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 11, pp. 16196–16210, 2024.